

Executive Summary for Government Stakeholders

European Bank Customer Churn Predictions & Risk Scoring

Purpose

This research project develops an analytical workflow for identifying customers with different estimated probabilities of leaving a European bank. It uses 10,000 customer observations and combines data preparation, exploratory analysis, supervised machine learning, risk scoring, threshold analysis, what-if scenario testing and an interactive Streamlit application. The intended value is to support evidence-based customer-retention analysis while recognizing that predictive results do not establish causality.

Key Evidence from the Analysis

- The dataset contains 10,000 customers, with 2,037 recorded as churned and an overall churn rate of 20.37%.
- The modelling workflow used a stratified 80/20 split, producing 8,000 training records and 2000 test records.
- Gradient Boosting recorded the highest test ROC-AUC in the research paper at 0.8692. Random Forest recorded ROC-AUC of 0.8651, recall of 0.7150 and F1-score of 0.6198 and was subsequently used for risk scoring and explainability.
- Random Forest churn probabilities ranged from 0.0224 to 0.9780 and were grouped into Low Risk(<0.30), Medium Risk (0.30 to <0.60) and High Risk(>=0.60).
- Observed churn rates differed across geography, engagement status, gender and age groups. Germany recorded 32.44% churn, compared with 16.15% in France and 16.67% in Spain.
- Inactive members recorded 26.85% churn compared with 14.27% for active members. These are associations in the supplied dataset and should not be interpreted as casual effects.

Relevance for Government and Public-Sector Stakeholders

The project illustrates how customer-level predictive analytics can function as a decision-support mechanism in financial services. Risk scores can help organizations prioritize analytical review and retention resources rather than applying identical interventions to every customer. It also demonstrates why governance is important when predictive models are used in banking. The research identifies calibration, subgroup performance, model drift, threshold selection, external validation and responsible interpretation as important considerations before operational deployment.

Operational Risk and Threshold Considerations

The research shows a trade-off between false positives and false negatives. At a 0.50 threshold, the Random Forest produced 241 false positives and 116 false negatives. At a 0.70 threshold, false positives decreased to 61, while false negatives increased to 206. Threshold selection should therefore be aligned with the costs of unnecessary intervention, missed high-risk customers, retention value and operational capacity.

Explainability and Responsible Use

Age and number of products were the two highest Random Forest feature-importance variables in the reported model. The project also uses SHAP and partial-dependence analysis. These tools can support analyst review, but the paper treats them as model explanations rather than proof that changing an input will cause customer behaviour to change.

Recommended Governance Actions

- Require independent or temporal validation before using churn-risk scores for consequential operational decisions.
- Document the rationale for the selected predictions threshold and reassess it as costs and customer behaviour change.

- Monitor false positives, false negatives, realized churn, probability calibration, model drift and subgroup performance after deployment.
- Investigate segment difference without treating demographic or geographic variables as causal explanations.
- Validate unusually high product-count churn rates against group sizes and data quality before designing interventions.
- Use explainability outputs to support human review and governance rather than as a substitute for causal evidence.

Limitations Relevant to Policy Interpretation

The research does not establish that the dataset represents the wider banking population, and it does not provide temporal or external validation. It also does not establish probability calibration or a formal fairness audit. The what-if simulator provides predictive counterfactual outputs for individual records and does not establish causal effects. Findings should therefore be treated as evidence from the supplied dataset rather than as general policy conclusions.

Future Work

Future work proposed in the paper includes temporal and external validation, probability calibration, cost-sensitive threshold optimization, formal fairness and subgroup analysis, broader hyperparameter tuning, model monitoring, drift detection, model versioning , governance controls and controlled evaluation of retention interventions.

Executive Takeaway

The project demonstrates a complete predictive analytics pipeline that translates customer data into interpretable risk scores and an interactive decision-support interface. For government and regulatory stakeholders, the central governance lesson is the importance of combining predictive

performance with transparency, validation, threshold governance monitoring and responsible interpretation when machine-learning systems are used in banking-related decision support.

Source : European bank csv file and prepared using the research paper and Python , Pandas and Streamlit.